

REAL TIME PROJECT REPORT

BHOOMI – AI Powered Farming Assistant

Submitted By

Rajamuri Sai Rithik Reddy

B.Tech CSE (Data Science)

Prof. C. R. Rao AIMSCS

24XV1M6741

ABSTRACT

Agriculture is one of the most important sectors in India, but many farmers still face problems such as lack of proper guidance, pest attacks, incorrect fertilizer usage, and difficulty in selling their produce at fair prices. During my interactions with farmers in Telangana villages, I observed that many decisions were being made based on assumptions rather than verified information.

To address this problem, I started working on Bhoomi, an AI-powered farming assistant that provides farmers with agricultural guidance using Artificial Intelligence and verified agricultural data sources. The project aims to help farmers make better decisions regarding crop selection, fertilizer application, pest management, and market opportunities.

Bhoomi combines modern technologies such as Large Language Models (LLMs), Retrieval Augmented Generation (RAG), Python, and web technologies to provide a simple conversational platform for farmers. The long-term vision is to build a complete digital ecosystem connecting farmers, experts, buyers, and agricultural service providers.

1. INTRODUCTION

India is an agricultural nation where millions of families depend on farming for their livelihood. However, many farmers do not have immediate access to agricultural experts. This often results in lower yields, higher costs, and financial losses.

As a student and aspiring entrepreneur, I wanted to solve a real-world problem rather than build a theoretical academic project. This led to the development of Bhoomi.

The idea originated when I noticed that farmers often had the same questions repeatedly:

- Which crop should I grow?
- Which fertilizer should I use?
- How do I control pests?
- Where can I sell my produce?

Most farmers either depend on local dealers or search for information from multiple sources. Bhoomi was designed to provide accurate answers in one place.

2. OBJECTIVES OF THE PROJECT

The primary objectives of Bhoomi are:

1. To provide instant agricultural guidance to farmers.
 2. To help farmers choose suitable crops based on local conditions.
 3. To reduce crop losses caused by pests and diseases.
 4. To connect farmers with potential buyers.
 5. To promote data-driven farming practices.
 6. To make agricultural knowledge easily accessible through AI.
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3. PROBLEM IDENTIFICATION

During discussions with farmers, several challenges were identified:

Lack of Agricultural Knowledge

Many farmers rely on traditional methods without access to updated agricultural information.

Pest and Disease Management

Farmers often fail to identify crop diseases at an early stage.

Market Challenges

Farmers frequently sell products through intermediaries, reducing their profits.

Information Gap

Reliable agricultural information is scattered across multiple websites and government portals, making it difficult for farmers to access.

These challenges motivated me to build Bhoomi as a centralized digital solution.

4. PROJECT DEVELOPMENT

Planning Phase

Initially, I researched existing agricultural applications and identified their limitations. Most platforms either provided static information or were difficult for farmers to use.

I then decided to build a conversational AI platform where farmers could ask questions naturally.

Technology Used

Programming Language

- Python

Frontend

- HTML
- CSS
- JavaScript

AI Technologies

- Large Language Models (LLMs)
- Retrieval Augmented Generation (RAG)

Database

- Agricultural datasets collected from trusted sources

Development Tools

- VS Code
- GitHub
- Streamlit

5. SYSTEM WORKING

The working process of Bhoomi is simple.

Step 1

The farmer enters a question.

Example:

"What fertilizer should I use for paddy cultivation?"

Step 2

The RAG system retrieves relevant information from agricultural documents and databases.

Step 3

The AI model processes the information and generates a contextual response.

Step 4

The farmer receives recommendations in simple language.

This process helps reduce misinformation and improves decision-making.

6. CURRENT PROGRESS

The project is currently in the prototype stage.

Achievements so far include:

- Developed initial AI assistant.
- Tested agricultural question-answering system.
- Connected with around 10 farmers.
- Covered approximately 20 acres through farmer interactions.
- Collected feedback regarding farming challenges.
- Identified future marketplace opportunities.

The project is continuously being improved based on user feedback.

7. CHALLENGES FACED

While developing Bhoomi, I encountered several challenges.

Data Collection

Obtaining reliable agricultural data from trusted sources required significant research.

AI Response Accuracy

Ensuring that AI-generated responses are accurate and relevant was a major challenge.

Farmer Adoption

Many farmers are unfamiliar with AI technology, requiring awareness and demonstrations.

Resource Constraints

As a student project, development was carried out with limited financial resources.

Despite these challenges, the project has made steady progress.

8. LEARNING OUTCOMES

This project helped me gain practical experience in:

- Artificial Intelligence
- Large Language Models
- Retrieval Augmented Generation
- Full Stack Development
- Problem Solving
- Startup Development
- User Research
- Product Design
- Team Coordination

More importantly, it taught me how technology can be used to solve real social problems.

9. FUTURE SCOPE

The future scope of Bhoomi includes:

- Mobile Application Development
- Voice-based farmer interaction in regional languages
- Disease detection through image analysis
- Weather-based recommendations
- Equipment rental marketplace
- Direct farmer-to-buyer platform
- Integration with government agricultural schemes

The long-term goal is to create a complete digital ecosystem for Indian farmers.

CONCLUSION

Bhoomi is a real-time project developed with the objective of addressing practical problems faced by farmers. Unlike conventional academic projects, this initiative focuses on solving a real-world challenge through Artificial Intelligence and modern technology.

The project has already received positive feedback from farmers and has demonstrated the potential of AI in agriculture. As development continues, Bhoomi aims to contribute towards improving farmer productivity, profitability, and access to reliable agricultural knowledge.

This project has also provided me with valuable technical, entrepreneurial, and leadership experience as a student founder.